

Pratik Narendra Gupta

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SUMMARY OF QUALIFICATIONS

Research-driven Software Engineering student with proven experience designing and deploying intelligent, distributed, and embedded systems across aerospace, AI, and cloud environments. Strong foundation in systems architecture, machine learning, and real-time control, with hands-on expertise building end-to-end solutions from data pipelines and perception models to autonomous mission systems and firmware-level integrations. Demonstrated ability to lead complex, cross-functional engineering projects, including autonomous UAV systems, predictive maintenance platforms, and large-scale video analytics applications, with experience in simulation, Software-in-the-Loop (SIL), and Hardware-in-the-Loop (HIL) validation. Skilled in translating research into production-ready systems, with a focus on distributed architectures, telemetry systems, and data-driven decision-making, and passionate about building resilient, high-performance systems at the intersection of AI, robotics, and cloud-native infrastructure.

EDUCATION

Bachelor of Engineering Science, Software Engineering

2021-2026

Western University, London, Ontario

- Dean Honors list - 3rd year
- University of Western Ontario Distinction Scholarship (top 10%) 2021

WORK EXPERIENCE

Software Developer & Product Owner – Capstone Project

September 2025 - Present

Sponsored by Canada Basketball

- Architected an automated 3x3 basketball analysis system leveraging RF-DETR and SAM 2 for high-precision object detection and segmentation, utilizing ByteTrack and custom Vision-Language Models to maintain robust player identity and tracking across broadcast footage.
- Developing a sophisticated data processing layer that converts 2D video coordinates to 3D court positions via homography and synchronizes game events with millisecond-precision OCR clock reading to derive professional coaching insights such as shooting efficiency and pace factors.
- Developed a responsive, high-performance video analysis platform using Next.js, Tailwind CSS, and Docker, featuring a split-screen interface for real-time "Momentum Metrics" visualization and integrating secure Google OAuth authentication.

AI Engineer

September 2025 - Present

Motorshield – Western Cyber Society

- Architected an end-to-end predictive maintenance platform for motor telemetry, designing a scalable data pipeline (Simulink → ingestion services → TimescaleDB → ML models) to process high-frequency time-series sensor data (voltage, torque, RPM, temperature).
- Built data preprocessing pipeline (windowing, labeling, augmentation) converting raw signals into model-ready datasets with overlapping 40ms windows and fault classification.
- Developed and evaluated deep learning models (CNN, Transformer, Hybrid CNN–Transformer ~464k params) for multi-class fault detection using 6-channel time-series inputs.
- Implemented robust data augmentation strategies (noise injection, harmonics, drift, transient spikes) to improve real-world generalization across hardware variability.
- Awarded Industry Choice Award, Canadian Tech Summit — selected by industry judges for exceptional innovation and real-world impact beyond standard evaluation criteria.

Software lead

September 2025 - Present

Western Aero Design

- Architected an end-to-end autonomous mission system for a vtol platform, defining GNC workflows, mission sequencing, and service boundaries across flight and payload subsystems.
- Developed and customized PX4 firmware, including building custom drivers and bootloader modifications, enabling seamless integration of onboard sensors, payload systems, and communication interfaces.
- Designed and implemented low-latency, secure telemetry and command protocols (custom message schemas over LoRa), enabling real-time coordination between aircraft, payload, and ground station under constrained network conditions.
- Engineered a Raspberry Pi-based payload system, integrating motors, sensors, and embedded Linux services; delivered fault-tolerant autonomous deployment with reliable hardware–software interfacing (SPI/I2C/UART).
- Led System Integration (SIL) and Hardware-in-the-Loop (HIL) testing pipelines, driving iterative debugging, parameter tuning, and root-cause analysis to reduce mission-critical failures.
- Achieved 3rd place in Presentation and 4th place in Technical Report at the SAE Aero Design 2026 (Tampa, Florida),

demonstrating strong system design, documentation, and technical communication.

CO-OP Packaging Engineer Ajinomoto Foods North America (California)

May 2024 - April 2025

Ajinomoto Foods North America (California)

- Database Architecture & Migration: Led the consolidation of over 26,000 packaging data points into a new central database (Product Vision). Analyzed over 750 legacy specifications to design a streamlined schema, reducing attributes to 31 essentials through collaboration with R&D, Procurement, and New Product Introduction teams.
- Cost-Saving Analysis: Developed a Python-based tool using Pandas and NumPy to analyze case similarities with the Euclidean distance formula. Identified over \$150,000 in potential annual savings through corrugate case consolidation.
- Executive Presentation: Presented the cost-saving analysis findings to senior leadership, including the CEO and Chief Supply Chain Officer.
- Cross-Functional Data Support: Served as the primary data resource for major company initiatives, providing critical packaging data for 17 new robot palletizing systems and sustainability reporting to major clients like Kroger and US Foods.
- System Customization: Designed and configured a dynamic reporting template within the Product Vision system to automate the generation of standardized specification sheets for cross-functional teams.

VP of Project at Western Artificial Intelligence

2023 - 2024

- Leading a year-long program for students, enabling them to develop and deploy AI applications showcased at major conferences like CUCAI.
- Demonstrated exceptional time management skills, balancing multiple priorities to meet deadlines and deliver high-quality results.
- Enhancing team support and technical assistance to multiple coding teams for project development.

Developer at Western Artificial Intelligence

2022 - 2023

- Investigated motor imagery EEG signal processing using machine learning techniques on open-source datasets.
- Modeled neural activity using signal processing and classification pipelines to study brain-computer interface (BCI) applications.
- Represented Western University at CUCAI (Canadian Undergraduate Conference on Artificial Intelligence)
- Schultz, L., Gupta, P., Choudhury, S. "Classifying Motor Imagery from Open-Source Datasets." Proceedings of the Canadian Undergraduate Conference on Artificial Intelligence (CUCAI), 2023, p.
- <https://drive.google.com/file/d/16ZJLhA6t23tT-D9DQtLmKx-DZMM-XeW/view?usp=sharing>

PROJECT EXPERIENCE

Home Server

2023 - present

- Architected a scalable and secure home server utilizing Docker for containerization, with Traefik as a dynamic reverse proxy to automatically manage network routing and SSL certificate acquisition for all hosted services.
- Engineered a multi-layered security framework by leveraging Cloudflare for DNS management and protection against DDoS attacks, significantly enhancing the server's public-facing resilience.
- Integrated Google OAuth into the reverse proxy middleware to enforce secure, single sign-on (SSO) authentication for internal services, providing granular access control without relying on a traditional VPN.
- Deployed CrowdSec, an automated intrusion prevention system, to proactively analyze traffic, detect malicious behavior, and block threats in real-time across the network.
- Created a custom Next.js dashboard to monitor and manage server services, providing a user-friendly interface for visualizing crucial statistics from an integrated MySQL database.
- Experimental platform to study system reliability, authentication boundaries, and fault isolation in distributed services.

Customer Purchase Recommendation System

May 2024 – April 2025

- Developed a Python-based recommendation tool using TensorFlow, Pandas, NumPy, and Scikit-Learn, achieving 96% accuracy in predicting customer purchases.
- Designed an intuitive user interface with Next.js and deployed the system on an AWS EC2 instance for scalable performance.

SKILLS

ML & AI: Predictive Maintenance, CNNs, LSTM/GRU, Transformers, Segmentation , Feature Engineering, Data Augmentation

Programming Languages: Python, Java, JavaScript and BASH

Cloud and Development Tools: PyTorch, TensorFlow, Docker, Kubernetes, AWS, Git

Embedded & Robotics: PX4, Raspberry Pi, Embedded Linux, SPI/I2C/UART, Real-Time Systems, Hardware-Software Integration